

Name

Class

Date

Scientific method and variables review

Total questions: 24

Worksheet time: 14mins

1. The type of data that is non-numerical – descriptive
 - a) Quantitative data
 - b) Qualitative data
2. Data collected from an investigation that is numerical
 - a) Quantitative data
 - b) Qualitative data
3. After analyzing observations and data collected from an experiment, we must
 - a) Draw a conclusion
 - b) Ask a question
 - c) Research
 - d) Form a hypothesis
4. Hypotheses can be tested through
 - a) Experiments and public opinion
 - b) Popularity of the hypothesis
 - c) Experiments and Observations
 - d) Asking as many people as possible
5. Hypotheses must be CORRECT
 - a) True
 - b) False
6. The use of your five senses to gather information is called
 - a) Inferences
 - b) Observations
 - c) Variables
 - d) Technology
7. “If soil temperatures rise, then plant growth will increase.” This is an example of a _____.
 - a) Scientific Law
 - b) Hypothesis
 - c) Independent Variable
 - d) Dependent Variable

8. In scientific investigations, data is defined as
- a) Careful measurements or factual observations
 - b) The amount of energy used in your equipment
 - c) The use of your five senses
 - d) The variable that stays the same
9. Of the two groups in an experiment, this one undergoes all the steps of the experiment including the independent variable
- a) Control group
 - b) Experimental group
 - c) Popular group
 - d) Common group
10. Of the two groups in an experiment, this one undergoes all the steps of the experiment except the one being tested to allow for a comparison
- a) Control group
 - b) Experimental group
 - c) Popular group
 - d) Common group
11. The variable that is observed and measured during an experiment. It can be considered the 'effect'.
- a) Independent variable
 - b) Dependent variable
 - c) Constant variable
12. The variable that the investigator deliberately changes to observe what it may 'cause'.
- a) Independent variable
 - b) Dependent variable
 - c) Constant variable
13. In order to scientifically investigate a problem/question, it must be ____.
- a) Interesting to everyone
 - b) Testable
 - c) Based on your opinion
 - d) A question of what is right or wrong
14. A series of steps that are used to investigate a natural occurrence
- a) Observations
 - b) Inferences
 - c) Experiments
 - d) Scientific Method

15. Austin wants to know how different types of colored light affect plant growth? What is the DV?
- a) Types of plant growth
 - b) types of colored lights
 - c) colored lights on plant growth
 - d) plant growth
16. Zariah wants to know how different types of colored light affect plant growth? What is the IV?
- a) types of plant growth
 - b) types of colored lights
 - c) plant growth on colored lights
 - d) plant growth
17. What part of the experiment does this describe:
The same type of bean plant, the same amount of water, and the same soil were used.
- a) Constant variables
 - b) Control group
 - c) Levels of IV
 - d) Hypothesis
18. What is the correct formula to write a hypothesis?
- a) If the IV
then the DV will (take a guess)
 - b) If the IV is (pick a level of IV)
then the DV will (take a guess)
 - c) If the DV is (increase/decrease)
then the Level of the IV
 - d) The IV is (pick a level of IV)
The DV will (take a guess)

19.



Which of the following is a quantitative observation?

- a) The bird is very colorful.
- b) The bird has 4 different colors on him.
- c) The bird has many, many feathers.
- d) It is a bird.

20. How many (independent) variables can be tested in a controlled experiment?

- a) One
- b) No more than two
- c) Three
- d) As many as you want

21. Any knowledge gained using any of the five senses.

- a) Variables
- b) Inferences
- c) Observations

22. Is the following question testable?

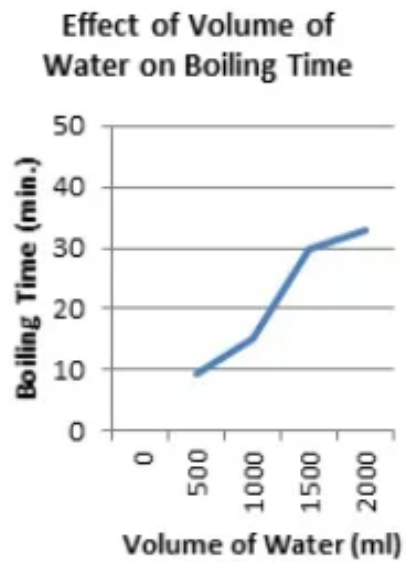
Are roses prettier flowers than tulips?

- a) yes
- b) no

23. What amount of sunlight makes pea plants grow tallest? What is the independent variable?

- a) height of pea plants
- b) amount of sunlight
- c) type of soil

24.



What conclusion can be drawn by using the information in the graph?

- a) The more water you have in a pot, the less time it takes to boil.
- b) The longer you boil water, the more water you get.
- c) The more water you have in a pot, the longer it takes to boil.